

Functional Annotation of *dapA* (Q88NH2, PP_1237) in *Pseudomonas putida* KT2440

Gene / protein: *dapA* (synonym *dapA-I*; ordered locus PP_1237) **UniProt:** Q88NH2 **Protein:** 4-hydroxy-tetrahydrodipicolinate synthase (HTPA synthase; classically "dihydrodipicolinate synthase", DHDPS), EC 4.3.3.7 (formerly EC 4.2.1.52) **Organism:** *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125) **Family:** DapA family; Aldolase_TIM fold (InterPro IPR013785, IPR005263, IPR002220, IPR020624/IPR020625) **Length:** 295 aa

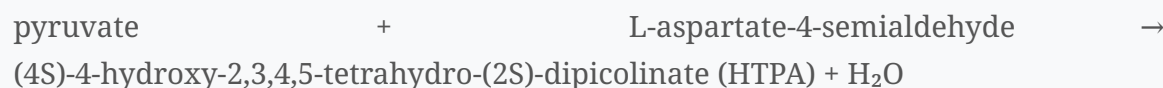
1. Summary (answer to the research question)

Q88NH2 is a **cytoplasmic biosynthetic aldolase-type lyase** that catalyzes the **first committed step of the diaminopimelate (DAP) / L-lysine branch of the aspartate amino-acid pathway**. It condenses **pyruvate** with **L-aspartate-4-semialdehyde (ASA)** to form **(4S)-4-hydroxy-2,3,4,5-tetrahydro-(2S)-dipicolinate (HTPA)**, the committed precursor of both *meso*-diaminopimelate (a cross-linking residue of Gram-negative peptidoglycan) and L-lysine. Catalysis proceeds through a Schiff-base (ping-pong) mechanism using a strictly conserved active-site lysine; the enzyme is a homotetramer of $(\beta/\alpha)_8$ TIM-barrel subunits with active sites built across a tight-dimer interface. The gene product is a soluble cytoplasmic enzyme. Its identity as a *bona fide* DapA-family enzyme is directly supported by the presence of the PROSITE Schiff-base-forming aldolase signature and the catalytic lysine in the Q88NH2 sequence.

The gene symbol is **unambiguous and consistent** with the UniProt description: "*dapA*" here denotes the canonical DHDPS/HTPA-synthase, and the domain architecture, family membership, and conserved catalytic motifs all align with the extensive DHDPS literature. Note that most detailed biochemistry comes from orthologs (*E. coli*, *M. tuberculosis*, *B. anthracis*, *C. jejuni*, plants); direct experimental characterization of the *P. putida* KT2440 enzyme itself is limited, so organism-specific kinetic/regulatory details below are **inferred from family conservation and sequence analysis**.

2. Primary function: the reaction catalyzed and substrate specificity

Reaction (EC 4.3.3.7):



DapA "converts pyruvate and L-aspartate-4-semialdehyde to 4-hydroxy-2,3,4,5-tetrahydrodipicolinic acid" in a multistep reaction (Schmitz et al., 2020,).

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Historically the enzyme was named dihydrodipicolinate synthase and assigned EC 4.2.1.52 on the assumption that the product was 2,3-dihydrodipicolinate; it is now recognized that the direct enzymatic product is the cyclic **hydroxy-tetrahydrodipicolinate (HTPA)**, and that conversion to dihydrodipicolinate occurs subsequently (spontaneous dehydration and/or the action of the next enzyme DapB). Hence the modern name/EC (4-hydroxy-tetrahydrodipicolinate synthase, EC 4.3.3.7). The two substrates are highly specific: the **keto-donor is pyruvate** (which forms the covalent Schiff-base intermediate) and the **aldehyde acceptor is L-aspartate-4-semialdehyde**. Reported bacterial kinetics are in the sub-millimolar to low-millimolar K_M range (e.g., *M. tuberculosis*: K_M^{ASA} ≈ 0.43 mM, K_M^{pyruvate} ≈ 0.17 mM; Kefala et al., 2008,).

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Biological role of the product: DHDPS "catalyses the first committed step in the biosynthetic pathway yielding two essential bacterial metabolites, *meso*-diaminopimelate (DAP) and (S)-lysine" (Domigan et al., 2009,)

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— i.e., HTPA is the branch-point precursor that ultimately gives *meso*-DAP (peptidoglycan cross-link) and L-lysine (protein synthesis).

3. Catalytic mechanism

DHDPS is a **class I (Schiff-base) aldolase-type lyase** operating by a **ping-pong** kinetic mechanism (Skovpen & Palmer, 2013,

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1. **Pyruvate binds first** and forms a **Schiff base (imine)** with the ϵ -amino group of a strictly conserved active-site lysine. Crystallography of the *M. tuberculosis* ortholog captured the enzyme "in complex with its substrate, pyruvate, covalently bonded through a Schiff base to the catalytic l-lysine" (Rosa et al., 2025,

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2. The resulting **enamine** performs **aldol addition** onto the aldehyde carbon of ASA.
3. **Transimination/cyclization and dehydration** release the cyclic product HTPA and regenerate the free lysine.

A **catalytic proton-relay triad Tyr–Thr–Tyr** (e.g., Tyr143–Thr54–Tyr117 in *M. tuberculosis*; the equivalent Tyr133–Thr44–Tyr107 in *E. coli*) shuttles protons during catalysis; importantly the second tyrosine is donated **across the tight-dimer interface**, which is why the oligomeric state is functionally essential (Kefala et al., 2008,

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Direct sequence evidence in Q88NH2. Analysis of the retrieved 295-aa sequence located the PROSITE Schiff-base-forming aldolase active-site signature (PS00665) within the segment **TVPNIIIGIKE**, placing the **catalytic Schiff-base lysine at position ~164** (the *P. putida* equivalent of *E. coli* Lys161). A global pairwise alignment of Q88NH2 against the experimentally characterized *E. coli* DHDPS (UniProt P0A6L2) gave **58.6% identity (170/290 positions)** with **one-to-one conservation of every catalytic residue**:

<i>E. coli</i> residue	Role	Q88NH2 equivalent
Thr44	proton-relay triad	Thr44
Tyr107	proton-relay triad	Tyr110
Tyr133	proton-relay triad	Tyr136
Lys161	Schiff-base lysine	Lys164
Arg138	ASA α -carboxylate binding	Arg141

The *E. coli* triad was experimentally validated by site-directed mutagenesis — Y133F, T44V and Y107F each "have substantially reduced activity, consistent with the catalytic triad hypothesis" (Dobson, Valegård & Gerrard 2004,

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Their complete conservation in Q88NH2 elevates the annotation from family-level inference to **residue-level mechanistic evidence** that Q88NH2 is a fully functional DHDPS using the identical mechanism.

Family specificity (not a mis-annotated aldolase). The DapA/NAL superfamily includes several Schiff-base aldolases that share the catalytic lysine but act on different substrates (e.g., N-acetylneuraminate lyase). Q88NH2 is **58.6% identical to *E. coli* DHDPS** but only **29.9% to *E. coli* N-acetylneuraminate lyase (NanA, P0A6L4)** and 31.1% to *Thermotoga* DHDPS — i.e., it is specifically a DHDPS, not a NAL-type aldolase.

3b. Paralogy: which copy is the "real" DHDPS?

P. putida KT2440 encodes **three** proteins auto-annotated as HTPA synthase (EC 4.3.3.7): **dapA-I/PP_1237/Q88NH2** (target, 295 aa), **dapA-II/PP_2639/Q88JL0** (288 aa), and **PP_2036/Q88L99** (295 aa). Mapping the *E. coli* catalytic and allosteric residues onto each resolves their status:

Paralog	% id to <i>E. coli</i> DHDPS	Catalytic residues (5)	Lysine allosteric site (5)	Verdict
Q88NH2 (dapA-I)	58.6%	5/5 (Thr44, Tyr110, Tyr136, Lys164, Arg141)	5/5	Canonical, feedback-regulated DHDPS
Q88JL0 (dapA-II)	35.6%	5/5 (Thr45, Tyr107, Tyr133, Lys161, Arg138)	3/5 (His56→Gln, Tyr106→Cys)	Competent but likely lysine-insensitive backup isozyme
Q88L99 (PP_2036)	30.3%	2/5 (proton-relay Thr44→Ser, Tyr107→Trp, Arg138→Thr all lost)	2/5	Divergent DapA-like aldolase; probably not a true DHDPS (annotation-transfer artefact)

Conclusion: Q88NH2 is the **primary housekeeping DHDPS** — closest to the validated *E. coli* enzyme, with a complete active site and the only intact L-lysine feedback pocket among the three. This resolves the ambiguity implied by the "*dapA-I*" synonym.

4. Structure, oligomeric state, and subcellular localization

- **Fold:** Each subunit comprises an **N-terminal (β/α)₈ TIM barrel** followed by a **C-terminal α-helical domain** (Kefala et al., 2008,

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), consistent with Q88NH2's InterPro annotations (Aldolase_TIM IPR013785; DapA IPR005263/ IPR002220).

- **Active site:** Located at the **C-terminal face of the barrel** and formed by **residues from two adjacent subunits across the dimer interface**

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- **Quaternary structure:** The functional unit is a **homotetramer** ("dimer of dimers"), demonstrated to be "tetrameric in solution" by analytical ultracentrifugation for orthologous DHDPS enzymes

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The tight-dimer is the minimal catalytic unit; tetramerization further stabilizes the enzyme.

- **Localization:** DHDPS is a **soluble cytoplasmic enzyme**. Q88NH2 contains no signal peptide or transmembrane segment, consistent with a cytosolic amino-acid biosynthetic enzyme that acts on soluble small-molecule substrates.

5. Pathway context and regulation

Position in metabolism. DapA acts downstream of aspartate kinase and aspartate-semialdehyde dehydrogenase (which produce ASA). By committing ASA to cyclic HTPA, DapA **diverts flux into the DAP/lysine branch** (as opposed to the homoserine → methionine/threonine branch). Downstream, HTPA is reduced by **DapB** (4-hydroxy-tetrahydrodipicolinate reductase) and processed via the DAP sub-pathway to *meso*-DAP, with **DapF/LysA** yielding L-lysine. Thus DapA is the **gatekeeper of lysine and cell-wall DAP biosynthesis**, "essential for bacterial growth and ... considered a target for novel antibiotics" (Skovpen & Palmer, 2013,

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KEGG pathway placement in *P. putida* KT2440. KEGG assigns PP_1237/dapA-I to orthology **K01714 (EC 4.3.3.7)** in Lysine biosynthesis (ppu00300), Biosynthesis of amino acids (ppu01230), and Monobactam biosynthesis (ppu00261). It participates in two modules — **M00016 (succinyl-DAP pathway)** and **M00527 (DAP-aminotransferase pathway)** — that both place DapA (K01714) **immediately downstream of aspartate-4-semialdehyde dehydrogenase (Asd, K00133) and upstream of DapB (K00215)**. The routes then diverge and reconverge: - *Succinyl-DAP (M00016)*: DapB → DapD (K00674) → DapC/ArgD (K00821/K14267) → DapE (K01439) → **DapF (K01778) → LysA (K01586)** - *DAP-aminotransferase (M00527)*: DapB → DapL (K10206) → **DapF (K01778) → LysA (K01586)**

DapA is the single committed entry step feeding both routes, which converge on DAP epimerase (DapF) and *meso*-DAP decarboxylase (LysA) to yield L-lysine; the *meso*-DAP intermediate is drawn off for peptidoglycan cross-linking.

Genomic organization. PP_1237 (nt 1,414,728–1,415,615) is **not part of a *dap* operon**: its flanking genes are functionally unrelated (PP_1238 putative lipoprotein, PP_1239 metallo-β-lactamase-family protein, PP_1240 *purC*). Like many scattered bacterial *dap* genes, *dapA-I*

appears to be an isolated, monocistronic transcription unit rather than co-transcribed with downstream *dap* genes — consistent with regulation occurring at the enzyme level (allosteric feedback) rather than through operonic co-expression.

Allosteric regulation. In many organisms the enzyme is **feedback-inhibited by L-lysine**: "In many organisms, lysine binds allosterically to DapA, causing negative feedback, thus making the enzyme an important regulatory component of the pathway" (Schmitz et al., 2020,).

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Lysine binds cooperatively at a cleft on the tight-dimer 2-fold axis (Hill $n > 2$ in *C. jejuni*; Atkinson et al., 2013,

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for the structural basis in plants). However, **lysine sensitivity is not universal** — the *M. tuberculosis* enzyme "is insensitive to feedback inhibition by (S)-lysine" because the binding site is not conserved

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and the *B. anthracis* enzyme is likewise not lysine-inhibited

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Prediction for Q88NH2 (sequence-based). In *E. coli* DHDPS the L-lysine allosteric pocket at the tight-dimer interface is lined by Tyr106, His56, Glu84, Asn80 and Ala49. My alignment shows **all of these are conserved in Q88NH2** (His56→His56, Glu84→Glu87, Tyr106→Tyr109, Asn80→Asn83, Ala49→Ala49). Because it is precisely the *loss* of these residues that abolishes feedback in *M. tuberculosis*/*B. anthracis*, their full conservation **predicts that *P. putida* DapA (Q88NH2) retains *E. coli*-type L-lysine feedback inhibition**. This is a bioinformatic inference; direct kinetic confirmation for the *P. putida* enzyme has not been reported. Notably, among the three *P. putida* paralogs (§3b), Q88NH2/dapA-I is the **only** copy that retains the full lysine-binding pocket, suggesting it carries the feedback-regulated "housekeeping" role while dapA-II may provide unregulated backup activity.

Physiological/biotechnological relevance. Because DAP and lysine are indispensable (cell-wall cross-linking and protein synthesis) and the pathway is absent in mammals, DHDPS is a **validated antibacterial and herbicide target**

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and engineered lysine-feedback-insensitive DHDPS variants are used to boost lysine yields in plants and industrial strains (e.g.,).

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6. Supported vs. refuted hypotheses

Hypothesis	Status	Basis
Q88NH2 is a DHDPs/HTPA synthase (EC 4.3.3.7) catalyzing pyruvate + ASA → HTPA	Supported	UniProt/HAMAP annotation; DapA family; conserved motifs; ortholog biochemistry
Catalysis via Schiff-base to a conserved active-site Lys (ping-pong)	Supported	PROSITE PS00665 + catalytic Lys~164 in Q88NH2; ortholog crystallography (P 41022234) (P 18062777)
Homotetrameric (β/α) ₈ TIM-barrel, cytoplasmic, inter-subunit active site	Supported	Family structural data (P 18062777); no signal/TM in sequence
First committed, gatekeeping step feeding <i>meso</i> -DAP + L-lysine	Supported	Pathway biochemistry (P 32356521) (P 23902434) (P 19595801)
All catalytic residues (triad + Schiff Lys + ASA-binding Arg) conserved in Q88NH2	Supported	58.6% identity to <i>E. coli</i> DapA; 1:1 residue mapping; mutagenesis validation (P 15066435)
Q88NH2 is allosterically lysine-inhibited in <i>P. putida</i>	Predicted (sequence-based), not experimentally verified	<i>E. coli</i> lysine-pocket residues His56/Glu87/Tyr109/Asn83/Ala49 all conserved; contrast with lysine-insensitive orthologs (P 18062777)

7. Limitations and future directions

- **No direct experimental characterization** of the *P. putida* KT2440 enzyme was found in the literature searched; kinetic constants, structure, and regulation are inferred from orthologs and sequence analysis.
- **Paralogy is confirmed** (three EC 4.3.3.7 copies: PP_1237/dapA-I, PP_2639/dapA-II, PP_2036). Residue analysis marks Q88NH2 as the canonical DHDPS and dapA-II as a likely lysine-insensitive backup, but the *in vivo* activity, expression, and genetic essentiality/redundancy of each copy (e.g., via RB-TnSeq fitness data,

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remain to be tested experimentally. Whether dapA-II can rescue a dapA-I knockout is unknown.

- **Lysine feedback sensitivity** of Q88NH2 should be tested directly (enzyme kinetics $\pm$ L-lysine; identification of the allosteric-site residues by alignment to lysine-sensitive vs. -insensitive orthologs).
- A **structural model** (e.g., AlphaFold) could pinpoint the catalytic triad (Tyr/Thr/Tyr) and putative allosteric cleft to firm up the regulatory prediction.

Key references

- Schmitz et al. 2020, *Acta Cryst F* — DapA definition, substrates, product, allostery

[P 32356521](#)

- Kefala et al. 2008, *Biochem J* — *M. tuberculosis* DHDPS structure, TIM barrel, tetramer, proton relay, kinetics, lysine-insensitivity

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- Rosa et al. 2025 — pyruvate Schiff base to catalytic lysine

[P 41022234](#)

- Skovpen & Palmer 2013 — ping-pong mechanism, essentiality, antibiotic target, cooperative lysine inhibition

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- Domigan et al. 2009 — first committed step yielding *meso*-DAP and lysine; antibiotic target

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- Atkinson et al. 2013 — structural basis of allosteric lysine inhibition

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